EXP : 6

```
# Import required libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create and train the Naïve Bayes model
model = GaussianNB()
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Display predictions
print("Predicted Values:")
print(y_pred)

print("\nActual Values:")
```

```
print(y_test)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy:", accuracy * 100, "%")
```

```
Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Accuracy: 97.77777777777777 %
```